Proposition 24: For any $\gamma > 0$,

$$\sup_{\widehat{f}:\Pi\to\mathbb{R}} \mathsf{dec}_\gamma(\mathcal{F}, \widehat{f}) = \sup_{\widehat{f}\in\mathrm{co}(\mathcal{F})} \mathsf{dec}_\gamma(\mathcal{F}, \widehat{f}).$$

4.7 Exercises

Exercise 8 (Posterior Sampling for Multi-Armed Bandits): Prove that for the standard multi-armed bandit,

$$\underline{\mathsf{dec}}_\gamma(\mathcal{F}) \lesssim \frac{|\Pi|}{\gamma},$$

by using the Posterior Sampling strategy (select p to be the action distribution induced by sampling $f \sim \mu$ and selecting π_f), and applying the decoupling lemma (Lemma 9). Recall that here, $\underline{\mathsf{dec}}_\gamma(\mathcal{F})$ is the "maxmin" version of the DEC (4.28).

Exercise 9: Prove Proposition 19.

Exercise 10: In this exercise, we will prove Proposition 24 as follows. First, show that the left-hand side is an upper bound on the right-hand side. For the other direction:

1. Prove that

$$\inf_{\widehat{f}\in\mathrm{co}(\mathcal{F})} \mathbb{E}_{f\sim\mu}\,\mathbb{E}_{\pi\sim p}(f(\pi) - \widehat{f}(\pi))^2 \le \inf_{\widehat{f}:\Pi\to\mathbb{R}} \mathbb{E}_{f\sim\mu}\,\mathbb{E}_{\pi\sim p}(f(\pi) - \widehat{f}(\pi))^2. \tag{4.31}$$

2. Use the Minimax Theorem (Lemma 41 in Appendix A.3) to conclude Proposition 24.

5. REINFORCEMENT LEARNING: BASICS

We now introduce the framework of *reinforcement learning*, which encompasses a rich set of dynamic, stateful decision making problems. Consider the task of repeated medical treatment assignment, depicted in Figure 4. To make the setting more realistic, it is natural to allow the decision-maker to apply 여러 단계의 전략 *multi-stage strategies* rather simple one-shot decisions such as "prescribe a painkiller." In principle, in the language of structured bandits, nothing is preventing us from having each decision π^t be a complex multi-stage treatment strategy that, at each stage, acts on the patient's dynamic state, which evolves as a function of the treatments at previous stages. As an example, intermediate actions of the type "if patient's blood pressure is above X then do Y" can form a decision tree that defines the complex strategy π^t. Methods from the previous lectures provide guarantees for such a setting, as long as we have a succinct model of expected rewards. What sets RL apart from structured bandits is the *additional information* about the intermediate state transitions and intermediate rewards. This information facilitates *credit assignment*, the mechanism for recognizing which of the actions led to the overall (composite) decision to be good or bad. This extra information can reduce what would otherwise be exponential sample complexity in terms of the number of stages, states, and actions in multi-stage decision making.

이전 stage들에서의 treatment에서 기인된 현재의 환자의 state (dynamic state)에다가 새로운 decision을 행함.

<Credit Assignment>

내려온 수많은 decision(action) 중에 "전체적인" 결과를 망친 놈이 누군지 recognize ⇒ Reduce Complexity (in # of stages states actions)

환경이 알려져 있을 때
Optimal decision을 효율적으로 compute

① RL framework, Bellman optimality, DP

This section is structured as follows. We first present the formal reinforcement learning framework and present basic principles including Bellman optimality and dynamic programming, which facilitate efficiently computing optimal decisions when the environment is known. We then consider ② the case in which the environment is unknown, and ③ give algorithms for perhaps the simplest reinforcement learning setting, *tabular reinforcement learning*, where the state and action spaces are finite (and small). Algorithms for more complex reinforcement learning settings are given in Section 5.

② : 환경이 알려지지 X 케이스
③ : "tabular RL" 세팅에서의 알고리즘
↳ state & action spaces 유한

5.1 Finite-Horizon Episodic MDP Formulation

<MDP>
Each event depends SOLELY on the state attained in the previous event.

We consider an episodic finite-horizon reinforcement learning framework. With H denoting the *horizon*, a *Markov Decision Process* (MDP) M takes the form

$$M = \{\underbrace{\mathcal{S}}_{\text{State Space}}, \underbrace{\mathcal{A}}_{\text{Action Space}}, \{\underbrace{P_h^M}_{\text{Probability transition kernel}}\}_{h=1}^{\overset{\text{Horizon}}{H}}, \{\underbrace{R_h^M}_{\text{Reward Distribution}}\}_{h=1}^{\overset{\text{Horizon}}{H}}, \underbrace{d_1}_{\text{Initial State dist.}}\},$$

vary across MDP — fixed and known

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space,

$$P_h^M : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$$

is the probability transition kernel at step h,

$$R_h^M : \mathcal{S} \times \mathcal{A} \to \Delta(\mathbb{R})$$

is the reward distribution, and $d_1 \in \Delta(\mathcal{S})$ is the initial state distribution. We allow the reward distribution and transition kernel to vary across MDPs, but assume for simplicity that the initial state distribution is fixed and known.

For a fixed MDP M, an *episode* (회차!) proceeds under the following protocol. At the beginning of the episode, the learner ① selects a randomized, non-stationary *policy*

$$\pi = (\pi_1, \ldots, \pi_H),$$

where $\pi_h : \mathcal{S} \to \Delta(\mathcal{A})$; (we let Π_{rns} for "randomized, non-stationary" denote the set of all such policies.) ② The episode then evolves through the following process, beginning from State1 $\underset{\text{(initial state dist)}}{s_1 \sim d_1}$. For $h = 1, \ldots, \underset{\text{horizon}}{H}$: (In one episode)

- $\underset{\text{action}}{a_h} \sim \underset{\text{Policy (state)}}{\pi_h(s_h)}$.
- $\underset{\text{reward}}{r_h} \sim R_h^M(\underset{\text{state}}{s_h}, \underset{\text{action}}{a_h})$ and $\underset{\text{다음 state}}{s_{h+1}} \sim \underset{\text{Probability transition kernel}}{P_h^M}(s_h, a_h)$.

For notational convenience, we take s_{H+1} to be a deterministic terminal state. The *Markov property* refers to the fact that under this evolution,

Memoryless property.

$$\mathbb{P}^M(s_{h+1} = s' \mid s_h, a_h) = \mathbb{P}^M(s_{h+1} = s' \mid s_h, a_h, \overbrace{s_{h-1}, a_{h-1}, \ldots, s_1, a_1}^{\text{상관 X 라는의미!}}).$$

↳ Solely depend on the state at time h // independent of the state of the process at any time PREVIOUS to h.

The value for a policy π under M is given by

$$\underset{\text{Value for policy }\pi}{f^M(\pi)} := \underset{\text{기댓값}}{\mathbb{E}^{M,\pi}}\left[\sum_{h=1}^H \underset{\text{reward}}{r_h}\right], \tag{5.1}$$

where $\mathbb{E}^{M,\pi}[\cdot]$ denotes expectation under the process above. We define an optimal policy for model M as

A particular MDP may have multiple distinct optimal policies

$$\underset{\text{Optimal policy for model }M}{\pi_M} \in \arg\max_{\pi \in \Pi_{\mathsf{rns}}} f^M(\pi). \tag{5.2}$$

Optimal Policy
$\pi_M \in \arg\max_{\pi \in \Pi_{rns}} f^M(\pi)$

Value functions. Maximization in (5.2) is a daunting task, since each policy π is a complex multi-stage object. It is useful to define intermediate "reward-to-go" functions to start breaking this complex task into smaller sub-tasks. Specifically, for a given model M and policy π, we define the *state-action value function* and *state value function* via

(State-)action value function — 행동가치함수

State value function — 상태가치함수

$$Q_h^{M,\pi}(s,a) = \mathbb{E}^{M,\pi}\left[\sum_{h'=h}^{H} r_{h'} \mid s_h = s, a_h = a\right], \quad \text{and} \quad V_h^{M,\pi}(s) = \mathbb{E}^{M,\pi}\left[\sum_{h'=h}^{H} r_{h'} \mid s_h = s\right].$$

↳ 행동 가치 함수에 대해 평균값을 취한것

s부터 끝날때까지(H) reward 합! (G_t)
근데
$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots$ 처럼 discount factor γ를 곱해줌.
$V_\pi(s) = E_\pi[G_t \mid S_t = s]$ (정의)
$= E_\pi[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots \mid S_t = s]$
$= E_\pi[r_t + \gamma(r_{t+1} + \gamma r_{t+2} + \cdots) \mid S_t = s]$
$= E_\pi[r_t + \gamma(G_{t+1}) \mid S_t = s]$
$= E_\pi[r_t + \gamma V(S_{t+1}) \mid S_t = s]$

Hence, the definition in (5.1) reads

Value for policy π
$f^M(\pi) := \mathbb{E}^{M,\pi}\left[\sum_{h=1}^{H} r_h\right]$

$$f^M(\pi) = \mathbb{E}_{s\sim d_1, a\sim\pi_1(s)}\left[Q_1^{M,\pi}(s,a)\right] = \mathbb{E}_{s\sim d_1}\left[V_1^{M,\pi}(s)\right]$$

Online RL. For reinforcement learning, our main focus will be on what is called the *online reinforcement learning problem*, in which we interact with an unknown MDP $M^\star$ for T episodes. For each episode $t = 1, \ldots, T$, the learner selects a policy $\pi^t \in \Pi_{\mathsf{rns}}$. The policy is executed in the MDP $M^\star$, and the learner observes the resulting trajectory

resulting trajectory (on episode t)

$$\tau^t = (s_1^t, a_1^t, r_1^t), \ldots, (s_H^t, a_H^t, r_H^t).$$

1~H: 시간. (Horizon)
각 episode(회차)별로, 선택하는 policy를 다르게 함.
그렇게 여러 회차 볼장 다 볼때까지 (1~H) 시도해볼거고,
총 T개의 전략을 사용해볼거임. 즉, 총 T회차.

The goal is to minimize the total regret

Optimal policy — Policy on episode t

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[f^{M^\star}(\pi_{M^\star}) - f^{M^\star}(\pi^t)\right] \tag{5.3}$$

against the optimal policy $\pi_{M^\star}$ for $M^\star$.

The online RL framework is a strict generalization of (structured) bandits and contextual bandits (with i.i.d. contexts). Indeed, if $\mathcal{S} = \{s_0\}$ and $H = 1$, each episode amounts to choosing an action $a^t \in \mathcal{A}$ and observing a reward r^t with mean $f^M(a^t)$, which is precisely a bandit problem. On the other hand, taking $\mathcal{S} = \mathcal{X}$ and $H = 1$ puts us in the setting of contextual bandits, with d_1 being the distribution of contexts. In both cases, the notion of regret (5.3) coincides with the notion of regret in the respective setting.

We mention in passing that many alternative formulations for Markov decision processes and for the reinforcement learning problem appear throughout the literature. For example, MDPs can be studied with infinite horizon (with or without discounting), and an alternative to minimizing regret is to consider *PAC-RL* which aims to minimize the sub-optimality of a final output policy produced after exploring for T rounds.

5.2 Planning via Dynamic Programming

In some reinforcement learning problems, it is natural to assume that the true MDP $M^\star$ is known. This may be the case with games, such as chess or backgammon, where transition probabilities are postulated by the game itself. In other settings, such as robotics or medical treatment, the agent interacts with an unknown $M^\star$ and needs to learn at least some aspects of this environment. The online reinforcement learning problem described above falls in the latter category. Before attacking the learning problem, we need to understand the structure of solutions to (5.2) in the case where $M^\star$ is known to the decision-maker. In this section, we show that the problem of maximizing $f^M(\pi)$ over $\pi \in \Pi$ in a known model M (known as *planning*) can be solved efficiently via the principle of *dynamic programming*. Dynamic

transition probability가 미리 결정되어있는 게임들의 경우... $M^\star$ known

Online RL: True MDP $M^\star$가 unknown!!

Optimal policy 구하기

Ch2: Planning via DP

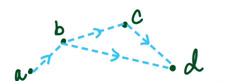
maximizing $f^M(\pi)$ (value) over $\pi \in \Pi$ in a KNOWN model M

Dynamic Programming: 복잡한 multi-Stage decision을 여러 개의 작은 decision들의 배열..로 쪼개어서 credit assignment 문제를 해결하는 것.

programming can be viewed as solving the problem of credit assignment by breaking down a complex multi-stage decision (policy) into a sequence of small decisions.

We start by observing that the optimal policy π_M in (5.2) may not be uniquely defined. For instance, if d_1 assigns zero probability to some state s_1, the behavior of π_M on this state is immaterial. (: 중요하지X(무관,하찮은)) In what follows, we introduce a fundamental result, Proposition 25, which guarantees existence of an optimal policy $\pi_M = (\pi_{M,1}, \ldots, \pi_{M,H})$ that maximizes $V_1^{M,\pi}(s)$ over $\pi \in \Pi_{\text{rns}}$ for all states $s \in \mathcal{S}$ *simultaneously* (rather than just on average, as in (5.2)). The fact that such a policy exists may seem magical at first, but it is rather straightforward. Indeed, if $\pi_{M,h}(s)$ (이런 policy..) is defined for all $s \in \mathcal{S}$ (..배열) and $h = 2, \ldots, H$, then defining the optimal $\pi_{M,1}(s)$ at any s is a matter of greedily choosing an action that maximizes the sum of the expected immediate reward and the remaining expected reward under the optimal policy. Indeed, this observation is Bellman's principle of optimality, stated more generally as follows [17]:

> **Principle of optimality.** *An optimal policy has the property that whatever the initial state and initial decisions are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decisions.*



만약 a-d의 최적 경로가 a-b-d라면 b-d는 노드 b부터 d까지의 최적 경로이다.

최적의 정책이라 함은,,! -이전의 (초기의) State와 decision이 무엇이었는지에 상관없이 남아 있는(그 이후의) 결정들은, 이전 결정으로 인해 초래된 상태에 대해 최적의 결정이다. (resulting from ... the first decisions)

최적의 정책은 초기의 상태와 결정이 무엇이었는지와 관계없이 남아있는 결정들이 초기의 결정들에 의해 초래된 상태에 대해 최적의 ~~결정~~ 정책이어야 한다는 특징을 가지고 있다

To state the result formally, we introduce the *optimal value functions* as

Optimal value functions (of each s, a, h)

Optimal (state-)action value function 최적 행동가치 함수 — State 최적 상태가치 함수

$$Q_h^{M,\star}(s,a) = \max_{\pi \in \Pi_{\text{rns}}} \mathbb{E}^{M,\pi}\left[\sum_{h'=h}^{H} r_{h'} \mid s_h = s, a_h = a\right] \quad \text{and} \quad V_h^{M,\star}(s) = \max_a Q_h^{M,\star}(s,a) \quad (5.4)$$

상태가치(State value)를 최대로 만드는 정책 π를 적용했을 때의 1) 행동가치 함수와 2) 상태가치 함수!

"최적"의 경우에는, 최적 상태가치 함수의 평균값이 아닌 "최댓값"을 취한 것.

for all $s \in \mathcal{S}$, $a \in \mathcal{A}$, and $h \in [H]$; we adopt the convention that "$V_{H+1}^{M,\star}(s) = Q_{H+1}^{M,\star}(s,a) = 0$." (terminal) Since these optimal values are separate maximizations for each s, a, h, it is reasonable to ask whether there exists a single policy that maximizes all these value functions simultaneously. Indeed, the following lemma shows that there exists $\boxed{\pi_M}$ such that for all s, a, h,

이 "각각"을 "동시에" 최대화하는 Single Policy가 존재하나? → OO!

$$Q_h^{M,\star}(s,a) = Q_h^{M,\boxed{\pi_M}}(s,a), \quad \text{and} \quad V_h^{M,\star}(s) = V_h^{M,\boxed{\pi_M}}(s). \quad \forall s,a,h \qquad (5.5)$$

Proposition 25 (Bellman Optimality): The optimal value function (5.4) for MDP M can be computed via $V_{H+1}^{M,\pi_M}(s) := 0$ (terminal), and for each $s \in \mathcal{S}$,

$$V_h^{M,\pi_M}(s) = \max_{a \in \mathcal{A}} \mathbb{E}^M\left[r_h + V_{h+1}^{M,\pi_M}(s_{h+1}) \mid s_h = s, a_h = a\right]. \qquad (5.6)$$

Optimal value function

r_h: h시점의 immediate reward! — $V_{h+1}^{M,\pi_M}(s_{h+1})$: future reward (에 discount factor가 곱해진...)

→ Compute the maximum over ALL actions!

즉, 현 시점의 State 가치함수가 (State value func.) 바로 다음 시점의 상태가치함수로 표현된다!

The optimal policy is given by

$$\pi_{M,h}(s) \in \arg\max_{a \in \mathcal{A}} \mathbb{E}^M\left[r_h + V_{h+1}^{M,\pi_M}(s_{h+1}) \mid s_h = s, a_h = a\right]. \qquad (5.7)$$

$\pi_{M,h}(s) \in \arg\max_{a \in A} V_h^{M,\pi_M}(s)$

Equivalently, for all $s \in \mathcal{S}$, $a \in \mathcal{A}$,

$$Q_h^{M,\pi_M}(s,a) = \mathbb{E}^M\left[r_h + \max_{a' \in \mathcal{A}} Q_{h+1}^{M,\pi_M}(s_{h+1}, a') \mid s_h = s, a_h = a\right], \qquad (5.8)$$

Compute the maximum over ALL actions!

and an the optimal policy is given by

$$\pi_{M,h}(s) \in \arg\max_{a\in\mathcal{A}} Q_h^{M,\pi_M}(s,a). \tag{5.9}$$

$Q_h^{M,\pi_M}(s,a) = \mathbb{E}^M\left[r_h + \max_{a'\in\mathcal{A}} Q_{h+1}^{M,\pi_M}(s_{h+1},a') \mid s_h=s, a_h=a\right]$

V_k로부터 V_{k+1}을 iteratively update until convergence to V^*

The update in (5.8) is referred to as *value iteration* (VI). It is useful to introduce a more succinct notation for this update. For an MDP M, define the *Bellman Operators* $\mathcal{T}_1^M, \ldots, \mathcal{T}_H^M$ via

vector space of state values

$\mathcal{T}^\pi : \mathbb{R}^n \to \mathbb{R}^n$ for any $v \in \mathbb{R}^n$

$\mathcal{T}^\pi = \mathcal{R}^\pi + \gamma P^\pi v$

operator: maps from one point to another within the vector space of state values, $\mathbb{R}^n$

$$[\mathcal{T}_h^M Q](s,a) = \mathbb{E}_{s_{h+1}\sim P_h^M(s,a), r_h\sim R_h^M(s,a)}\left[r_h(s,a) + \max_{a'\in\mathcal{A}} Q(s_{h+1},a')\right] \tag{5.10}$$

for any $Q : \mathcal{S}\times\mathcal{A}\to\mathbb{R}$. Going forward, we will write the expectation above more succinctly as

$$[\mathcal{T}_h^M Q](s,a) = \mathbb{E}^M\left[r_h(s_h,a_h) + \max_{a'\in\mathcal{A}} Q(s_{h+1},a') \mid s_h = s, a_h = a\right] \tag{5.11}$$

In the language of Bellman operators, (5.8) can be written as

$$Q_h^{M,\pi_M} = \mathcal{T}_h^M Q_{h+1}^{M,\pi_M}. \tag{5.12}$$

Bellman Operators

5.3 Failure of Uniform Exploration

ε-greedy를 online RL in MDP에서 사용 못하는 이유

The task of planning using dynamic programming—which requires knowledge of the MDP—is fairly straightforward, at least if we disregard the computational concerns. In this course, however, we are interested in the problem of learning to make decisions in the face of an unknown environment. Minimizing regret (in an unknown MDP) requires "exploration." As the next example shows, exploration in MDPs is a more delicate issue than in bandits.

Recall that ε-Greedy, a simple algorithm, is a reasonable solution for bandits and contextual bandits, albeit with a suboptimal rate ($T^{2/3}$ as opposed to $\sqrt{T}$). The next (classical) example, a so-called "combination lock," shows that such a strategy can be disastrous in reinforcement learning, as it leads to exponential (in the horizon H) regret.

Consider the MDP depicted in Figure 8, with $H+2$ states, and two actions a_g and a_b, and a starting state 1. The "good" action a_g deterministically leads to the next state in the chain, while the "bad" action deterministically leads to a terminal state. The only place where a non-zero reward can be received is the last state H, if the good action is chosen. The starting state is 1, and so the only way to receive non-zero reward is to select a_g for *all the H time steps* within the episode. Since the length of the episode is also H, selecting actions uniformly brings no information about the optimal sequence of actions, unless by chance all of the actions sampled happen to be good; the probability that this occurs is exponentially small in H. This means that T needs to be at least $O(2^H)$ to achieve nontrivial regret, and highlights the need for more strategic exploration.

근데 그럴거니가 ($1/2^H$)

ε-greedy NOT a good strategy in RL (leads to Exponential Regret)

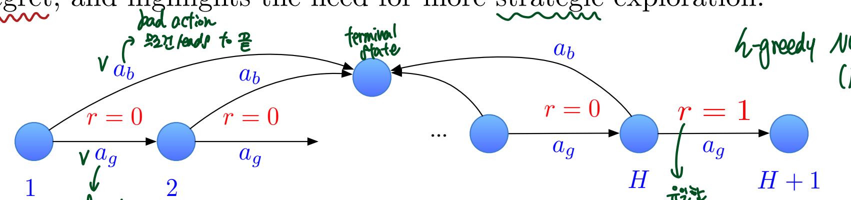


Figure 8: Combination Lock MDP.

Given the failure of ε-Greedy for this example, one can ask whether other algorithmic principles also fail. As we will show now, the principle of optimism succeeds, and an analogue of the UCB method yields a regret bound that is *polynomial* in the parameters $|\mathcal{S}|$, $|\mathcal{A}|$, and H. Before diving into the details, we present a collection of standard tools for analysis in MDPs, which will find use throughout the remainder of the lecture notes.

5.4 Analysis Tools

두 정책의 value의 차이를,, Single step decision에서의 차이로서 나타내는..!

One of the most basic tools employed in the analysis of reinforcement learning algorithms is the *performance difference lemma*, which expresses the difference in values for two policies in terms of differences in *single-step decisions* (made by the two policies). The simple proof, stated below, proceeds by successively changing one policy (π') into another (π) and keep track of the ensuing differences in expected rewards. One may also interpret this lemma as a version of the credit assignment mechanism. (전체-결과를 망친 "step"이 뭐였나~)

Henceforth, we adopt the following simplified notation. When a policy π is applied to the random variable s_h, we drop the subscript h and write $\pi(s_h)$ instead of $\pi_h(s_h)$, whenever this does not cause confusion.

Two Policies under Same MDP

Lemma 13 (Performance Difference Lemma): For any $s \in \mathcal{S}$, and $\pi, \pi' \in \Pi_{\mathsf{rns}}$,

differences in single-step decisions

$\pi(s_h)$

$$\underbrace{V_1^{M,\pi'}(s) - V_1^{M,\pi}(s)}_{\text{difference in values for two } (\pi \text{와 } \pi') \text{ policies}} = \sum_{h=1}^{H} \mathbb{E}^{M,\pi}\left[Q_h^{M,\pi'}(s_h, \pi'(s_h)) - Q_h^{M,\pi'}(s_h, a_h) \mid s_1 = s\right] \quad (5.13)$$

Proof of Lemma 13. Fix a pair of policies π, π' and define

1~H: 시간. (Horizon)
각 episode(회차)별로, 선택하는 policy를 다르게 함.
그렇게 여러 회차 볼 때 다 볼때까지 (1~H)
시도해볼거고,
총 T개의 전략을 사용해볼거임. 즉, 총 T회차.

$$\pi^h = (\pi_1, \ldots, \pi_{h-1}, \pi'_h, \ldots, \pi'_H),$$

$\pi^{\text{episode}}_{\text{horizon}}$

with $\pi^1 = \pi'$ and $\pi^H = \pi$. By telescoping, we can write

ep1: $\pi^1 = (\pi_1', \pi_2', \cdots, \pi'_H) = \pi'$
ep2: $\pi^2 = (\pi_1, \pi_2', \cdots, \pi'_H)$
ep3: $\pi^3 = (\pi_1, \pi_2, \pi_3', \cdots, \pi'_H)$
⋮ π'를 하나씩 π로 바꿔감 & keep track of the 그에따른 result
$\pi^H = (\pi_1, \cdots, \pi_H) = \pi$

$h=1$일때:

π^h 한 horizon step 더 π 선택

$$V_1^{M,\pi'}(s) - V_1^{M,\pi}(s) = \sum_{h=1}^{H} \left(V_1^{M,\pi^h}(s) - V_1^{M,\pi^{h+1}}(s)\right). \quad (5.14)$$

Observe that for each h, we have

by def. of V

$$\left(V_1^{M,\pi^h}(s) - V_1^{M,\pi^{h+1}}(s)\right) = \left(\mathbb{E}^{M,\pi^h}\left[\sum_{t=1}^{H} r_t \mid s_1 = s\right]\right) - \left(\mathbb{E}^{M,\pi^{h+1}}\left[\sum_{t=1}^{H} r_t \mid s_1 = s\right]\right). \quad (5.15)$$

Here, one process evolves according to (M, π^h) and the one evolves according to (M, π^{h+1}). The processes only differ in the action taken once the state s_h is reached. In the former, the action $\pi'(s_h)$ is taken, whereas in the latter it is $\pi(s_h)$. Hence, (5.15) is equal to

"s_h" state에 갔을 때만 r에 차이가 생김!

$$\mathbb{E}_{s_h \sim (M,\pi)} \mathbb{E}^{M,\pi}\left[Q_h^{M,\pi'}(s_h, \pi'(s_h)) - Q_h^{M,\pi'}(s_h, \pi(s_h)) \mid s_1 = s\right] \quad (5.16)$$

action value function의 form

which can be written as

$$\mathbb{E}_{(s_h, a_h) \sim (M,\pi)} \mathbb{E}^{M,\pi}\left[Q_h^{M,\pi'}(s_h, \pi'(s_h)) - Q_h^{M,\pi'}(s_h, a_h) \mid s_1 = s\right]. \quad (5.17)$$

□

In contrast to the performance difference lemma, which relates the values of two policies under the same MDP, the next result relates the performance of the same policy under two different MDPs. Specifically, the difference in initial value for two MDPs is decomposed into a sum of errors between layer-wise value functions.

Performance of "Same Policy" under two different MDPs. (첫번째파트만)

Lemma 14 (Bellman residual decomposition): For any pair of MDPs $M = (P^M, R^M)$ and $\widehat{M} = (P^{\widehat{M}}, R^{\widehat{M}})$, for any $s \in \mathcal{S}$, and policies $\pi \in \Pi_{\mathrm{RNS}}$,

State Value function

$$V_1^{M,\pi}(s) - V^{\widehat{M},\pi}(s) = \sum_{h=1}^{H} \mathbb{E}^{\widehat{M},\pi}\left[Q_h^{M,\pi}(s_h, a_h) - r_h - V_{h+1}^{M,\pi}(s_{h+1}) \mid s_1 = s\right] \tag{5.18}$$

error

Hence, for $M, \widehat{M}$ with the same initial state distribution,

Value for policy π

$$f^M(\pi) - f^{\widehat{M}}(\pi) = \sum_{h=1}^{H} \mathbb{E}^{\widehat{M},\pi}\left[Q_h^{M,\pi}(s_h, a_h) - r_h - V_{h+1}^{M,\pi}(s_{h+1})\right]. \tag{5.19}$$

same MDP

In addition, for any MDP M and function $Q = (Q_1, \ldots, Q_H, Q_{H+1})$ with $Q_{H+1} \equiv 0$, letting $\pi_{Q,h}(s) = \arg\max_{a \in \mathcal{A}} Q_h(s, a)$ (policy), we have

$$\max_{a \in \mathcal{A}} Q_1(s, a) - V_1^{M,\pi_Q}(s) = \sum_{h=1}^{H} \mathbb{E}^{M,\pi_Q}\left[Q_h(s_h, a_h) - [\mathcal{T}_h^M Q_{h+1}](s_h, a_h) \mid s_1 = s\right]. \tag{5.20}$$

and, hence,

$$\mathbb{E}_{s_1 \sim d_1}\left[\max_{a \in \mathcal{A}} Q_1(s_1, a)\right] - f^M(\pi_Q) = \sum_{h=1}^{H} \mathbb{E}^{M,\pi_Q}\left[Q_h(s_h, a_h) - [\mathcal{T}_h^M Q_{h+1}](s_h, a_h)\right]. \tag{5.21}$$

Note that for the second part of Lemma 14 $Q = (Q_1, \ldots, Q_H)$ can be any sequence of functions, and need not be a value function corresponding to a particular policy or MDP. (It is worth noting that Q gives rise to the greedy policy π_Q, which, in turn, gives rise to Q^{M,π_Q} (the value of π_Q in model M), but it may well be the case that $Q^{M,\pi_Q} \neq Q$.)

Proof of Lemma 14. We will prove (5.19), and omit the proof for (5.18), which is similar but more verbose. We have

(5.19의 proof)
(RHS을 먼저 만들어놓고 그걸 LHS 꼴로 변환해감.)

r_h는 밖으로

$$\begin{aligned}
\sum_{h=1}^{H} \mathbb{E}^{\widehat{M},\pi}\left[Q_h^{M,\pi}(s_h, a_h) - r_h - V_{h+1}^{M,\pi}(s_{h+1})\right] &= \sum_{h=1}^{H} \mathbb{E}^{\widehat{M},\pi}\left[Q_h^{M,\pi}(s_h, a_h) - V_{h+1}^{M,\pi}(s_{h+1})\right] - \mathbb{E}^{\widehat{M},\pi}\left[\sum_{h=1}^{H} r_h\right] \\
&= \sum_{h=1}^{H} \mathbb{E}^{\widehat{M},\pi}\left[Q_h^{M,\pi}(s_h, a_h) - V_{h+1}^{M,\pi}(s_{h+1})\right] - f^{\widehat{M}}(\pi).
\end{aligned}$$

f의 정의에 의해

$$V_h^{M,\pi}(s) = \mathbb{E}_{a \sim \pi_h(s)}\left[Q_h^{M,\pi}(s,a)\right]$$

$$\begin{aligned}
&= \sum_{h=1}^{H} \mathbb{E}^{\widehat{M},\pi}\left[V_h^{M,\pi}(s_h) - V_{h+1}^{M,\pi}(s_{h+1})\right] \\
&= \mathbb{E}^{\widehat{M},\pi}\left[V_1^{M,\pi}(s_1)\right] - \mathbb{E}^{\widehat{M},\pi}\left[V_{H+1}^{M,\pi}(s_{H+1})\right] \\
&= f^M(\pi)
\end{aligned}$$

On the other hand, since $V_h^{M,\pi}(s) = \mathbb{E}_{a\sim\pi_h(s)}[Q_h^{M,\pi}(s,a)]$, a telescoping argument yields

$$\begin{aligned}\sum_{h=1}^{H} \mathbb{E}^{\widehat{M},\pi}\left[Q_h^{M,\pi}(s_h,a_h) - V_{h+1}^{M,\pi}(s_{h+1})\right] &= \sum_{h=1}^{H} \mathbb{E}^{\widehat{M},\pi}\left[V_h^{M,\pi}(s_h) - V_{h+1}^{M,\pi}(s_{h+1})\right] \\ &= \mathbb{E}^{\widehat{M},\pi}\left[V_1^{M,\pi}(s_1)\right] - \mathbb{E}^{\widehat{M},\pi}\left[V_{H+1}^{M,\pi}(s_{H+1})\right] \\ &= f^M(\pi),\end{aligned}$$

$*V_{H+1}^{M,\pi} = 0.$

where we have used that $V_{H+1}^{M,\pi} = 0$, and that both MDPs have the same initial state distribution. We prove (5.21) (omitting the proof of (5.20)) using a similar argument. We have

RHS

$$\begin{aligned}&\sum_{h=1}^{H} \mathbb{E}^{M,\pi_Q}\left[Q_h(s_h,a_h) - r_h - \max_{a\in\mathcal{A}} Q_{h+1}(s_{h+1},a)\right] \\ &= \sum_{h=1}^{H} \mathbb{E}^{M,\pi_Q}\left[Q_h(s_h,a_h) - \max_{a\in\mathcal{A}} Q_{h+1}(s_{h+1},a)\right] - \mathbb{E}^{M,\pi_Q}\left[\sum_{h=1}^{H} r_h\right] \\ &= \sum_{h=1}^{H} \mathbb{E}^{M,\pi_Q}\left[Q_h(s_h,a_h) - \max_{a\in\mathcal{A}} Q_{h+1}(s_{h+1},a)\right] - f^M(\pi_Q).\end{aligned}$$

r_h 따로 빼내기

by 정의

state s_{h+1}에서 가장 action value가 크게 된.. 그 value.

애초에 정의가 (state-) action value가 가장 크도록 하는 action이니까!

는! action이 a_{h+1}일때 였을 것임.

즉, $\max_{a\in A} Q_{h+1}(s_{h+1},a) = Q_{h+1}(s_{h+1},a_{h+1})$

Since $a_{h+1} = \pi_{Q,h}(s_{h+1}) = \arg\max_{a\in\mathcal{A}} Q_{h+1}(s_{h+1},a)$, we have

$$\mathbb{E}^{M,\pi_Q}\left[Q_h(s_h,a_h) - \max_{a\in\mathcal{A}} Q_{h+1}(s_{h+1},a)\right] = \mathbb{E}^{M,\pi_Q}\left[Q_h(s_h,a_h) - Q_{h+1}(s_{h+1},a_{h+1})\right],$$

and the result follows by telescoping.

$\sum \boxed{} = \mathbb{E}^{M,\pi_Q}[Q_1(s_1,a_1)] - \mathbb{E}^{M,\pi_Q}[\underbrace{Q_{H+1}}_{0}(s_{H+1},a_{H+1})]$

$= \mathbb{E}_{s_1\sim d_1}[\max_{a\in A} Q_1(s_1,a)]$

□

Another similar analysis tool for MDPs, the *simulation lemma*, is deferred to Section 6 (Lemma 23). This result can be proven as a consequence of Lemma 14.

5.5 Optimism

optimism in the face of uncertainty (ch2) : 모두의 width가 shrink 하는 confidence interval을 만들수만 있다면, Regret이 bound 됨.

To develop algorithms for regret minimization in unknown MDPs, we turn to the principle of optimism, which we have seen is successful in tackling multi-armed bandits and linear bandits (in small dimension). (Recall that for bandits, Lemma 7 gave a way to decompose the regret of optimistic algorithms into width of confidence intervals.) What is the analogue of Lemma 7 for MDPs? Thinking of optimistic estimates at the level of expected rewards for policies π is unwieldy, and we need to dig into the structure of these multi-stage decisions. In particular, the approach we employ is to construct a sequence of optimistic *value functions* $\overline{Q}_1, \ldots, \overline{Q}_H$ which are guaranteed to over-estimate the optimal value function $Q^{M,\star}$. (For multi-armed bandits, implementing optimism amounted to adding "bonuses," constructed from past data, to estimates (for the reward function).) We will construct optimistic value functions in a similar fashion. Before giving the construction, we introduce a technical lemma, which quantifies the error (in using such optimistic estimates) in terms of *Bellman residuals*; Bellman residuals measure self-consistency (of the optimistic estimates) under the application of the Bellman operator.

Optimal Value function $Q^{M,\star}$ 을 "과대평가" 하도록 보장되어있어야 한다. 뭐가? $\underbrace{\overline{Q}_1, \ldots, \overline{Q}_H}_{\text{Sequence of Optimistic Value functions.}}$ 들이!

$\overline{Q}_h - \mathcal{T}_h^M \overline{Q}_{h+1}$

optimistic estimates을 사용하는 데 있어 생기는 error를 양적화하는,! in terms of "Bellman Residuals"

$V^* - V^{\hat{\pi}}$

↳ measure "Self Consistency" of optimistic estimates (under the 활용 of Bellman operator)

Lemma 14
(In terms of Bellman Residuals)

Lemma 15 (Error decomposition for optimistic policies): Let $\{\bar{Q}_1, \ldots, \bar{Q}_H\}$ be a sequence of functions $\bar{Q}_h : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ with the property that for all (s, a),
(sequence of OPTIMISTIC Value functions)

① 과대평가하도록 보장됨!

$$\underbrace{Q_h^{M,\star}}_{\text{Optimal Value function}}(s, a) \leq \bar{Q}_h(s, a) \tag{5.22}$$

∴ "Optimistic" 하다는 뜻

and set $\bar{Q}_{H+1} \equiv 0$. Let $\widehat{\pi} = (\widehat{\pi}_1, \ldots, \widehat{\pi}_H)$ be such that $\widehat{\pi}_h(s) = \arg\max_a \bar{Q}_h(s, a)$. Then for all $s \in \mathcal{S}$,

optimistic func.

Bellman Residual이 "작아야" $\bar{Q}_h$가 "self consistent"

② 같은 MDP 아래 다른 policy

$$\underbrace{V_1^{M,\star}(s) - V_1^{M,\widehat{\pi}}(s)}_{\text{errors... in using optimistic estimates}} \leq \sum_{h=1}^{H} \mathbb{E}^{M,\widehat{\pi}}\big[(\underbrace{\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1}}_{\text{Bellman Residual}})(s_h, \widehat{\pi}(s_h)) \mid s_1 = s\big]. \tag{5.23}$$

가 quantify 됨. In terms of Bellman Residuals!

$Q_h^{M,\pi_M} = \mathcal{T}_h^M Q_{h+1}^{M,\pi_M}$ (5.12)

RHS↓ implies LHS↓ ← Lemma 15 tells us that closeness of $\bar{Q}_h$ to the Bellman backup $\mathcal{T}_h^M \bar{Q}_{h+1}$ implies closeness of $\widehat{\pi}$ to π_M (in terms of the value). (As a sanity check, if $\bar{Q}_h = Q_h^{M,\star}$, the right-hand side of (5.23) is zero, since $Q_h^{M,\star} = \mathcal{T}_h^M Q_{h+1}^{M,\star}$.) Crucially, errors do not accumulate too fast as a function of the horizon. This fact should not be taken for granted: in general, if $\bar{Q}$ is not optimistic, it could have been the case that small changes in $\bar{Q}_h$ exponentially degrade the quality of the policy $\widehat{\pi}$.

Another important aspect of the decomposition (5.23) is the *on-policy* nature of the terms in the sum. Observe that the law of s_h for each of the terms is given by executing $\widehat{\pi}$ in model M. The distribution of s_h is often referred to as the *roll-in distribution*; when this distribution is induced by the policy executed by the algorithm we may have a better control of the error than in the *off-policy* case when the roll-in distribution is given by π_M or another unknown policy.

Proof of Lemma 15. Let $\bar{V}_h(s) := \max_{a \in \mathcal{A}} \bar{Q}_h(s, a)$. Just as in the proof of Lemma 7, the assumption that $\bar{Q}_h$ is "optimistic" implies that

과대 평가 해야 함! 정의 $\widehat{\pi}_h(s) = \arg\max_a \bar{Q}_h(s,a)$

$$Q_h^{M,\star}(s_h, \pi_M(s_h)) \leq \bar{Q}_h(s_h, \pi_M(s_h)) \leq \bar{Q}_h(s_h, \widehat{\pi}(s_h))$$

and, hence, $V_1^{M,\star}(s) \leq \bar{V}_1(s)$. Then, (5.20) applied to $Q = \bar{Q}$ and $\pi_Q = \widehat{\pi}$ states that

$$\bar{V}_1(s) - V_1^{M,\widehat{\pi}}(s) = \sum_{h=1}^{H} \mathbb{E}^{M,\widehat{\pi}}\big[\bar{Q}_h(s_h, a_h) - [\mathcal{T}_h^M \bar{Q}_{h+1}](s_h, a_h) \mid s_1 = s\big]. \tag{5.24}$$

(5.20)

$$\max_{a \in \mathcal{A}} Q_1(s, a) - V_1^{M,\pi_Q}(s) = \sum_{h=1}^{H} \mathbb{E}^{M,\pi_Q}\big[Q_h(s_h, a_h) - [\mathcal{T}_h^M Q_{h+1}](s_h, a_h) \mid s_1 = s\big]. \tag{5.20}$$

□

Remark 16: In fact, the proof of Lemma 15 only uses that the initial value $\bar{Q}_1$ is optimistic. However, to construct a value function with this property, the algorithms we consider will proceed by backwards induction, producing optimistic estimates $\bar{Q}_1, \ldots, \bar{Q}_H$ in the process.

5.6 The UCB-VI Algorithm for Tabular MDPs

Principle of Optimism을 활용해
tabular MDP 에서의 ← State & Action spaces가 작음.
online RL에 대한
regret bound를 계산! ← S, A, H에 대해 polynomial!

We now instantiate the principle of optimism to give regret bounds for online reinforcement learning in *tabular MDPs*. Tabular RL may be thought of as an analogue of finite-armed bandits: we assume no structure across states and actions, but require that the state and action spaces are small (finite!). The regret bounds we present will depend polynomially on $S = |\mathcal{S}|$ and $A = |\mathcal{A}|$, as well as the horizon H.

Reward func: 알려져 있음. 모르는 것: Transition 확률!

Preliminaries. For simplicity, we assume that the reward function is known to the learner, so that only the transition probabilities are unknown. This does not change the difficulty of the problem in a meaningful way, but allows us to keep notation light.

(for simplicity, Reward is known & $V \in [0,1]$)

Assumption 6: Rewards are deterministic, bounded, and known to the learner: $R_h^M(s,a) = \delta_{r_h(s,a)}$ for known $r_h : \mathcal{S} \times \mathcal{A} \to [0,1]$, for all M. In addition, assume for simplicity that $V_1^{M,\star}(s) \in [0,1]$ for any $s \in \mathcal{S}$.

Define, with a slight abuse of notation,

n: frequency

$\pi^{\text{episode}}_{\text{horizon}}$

$$n_h^t(\underbrace{s}_{\text{state}},\underbrace{a}_{\text{action}}) = \underbrace{\sum_{i=1}^{t-1}}_{\text{각 episode별 (in t-1) ...빈도수 셈.}} \mathbb{I}\left\{(s_h^i, a_h^i) = (s,a)\right\}, \quad \text{and} \quad n_h^t(\underbrace{s}_{\text{state}},\underbrace{a}_{\text{action}},\underbrace{s'}_{\text{next state}}) = \sum_{i=1}^{t-1} \mathbb{I}\left\{(s_h^i, a_h^i, s_{h+1}^i) = (s,a,s')\right\},$$

as the empirical state-action and state-action-next state frequencies (빈도수). We can estimate the transition probabilities via

transition 확률에 대한 측정치! (경험적)

transition prob. (emperical!)

$$\widehat{P}_h^t(s' \mid s,a) = \frac{n_h^t(s,a,s')}{n_h^t(s,a)}. \tag{5.25}$$

The UCB-VI algorithm. The following algorithm, UCB-VI ("Upper Confidence Bound Value Iteration") [16], combines the notion of optimism with dynamic programming.

> UCB-VI Upper Confidence Bound Value Iteration
> **for** $t = 1, \ldots, T$ **do** t: episode
> Let $\overline{V}_{H+1}^t \equiv 1$. terminal (Horizon)
> **for** $h = H, \ldots, 1$ **do** horizon (시간..)
> Update $\underbrace{n_h^t(s,a)}_{\text{각 빈도수.}}$, $n_h^t(s,a,s')$, and $b_{h,\delta}^t(s,a)$, for all $(s,a) \in \mathcal{S} \times \mathcal{A}$.
> `// `$b_{h,\delta}^t(s,a)$` is a bonus computed in (5.27).`
> Compute:
> (State-) action value func.
> $$\overline{Q}_h^t(s,a) = \left\{\underbrace{r_h(s,a)}_{\text{immediate reward}} + \overbrace{\mathbb{E}_{s' \sim \widehat{P}_h^t(\cdot|s,a)} \overline{V}_{h+1}^t(s')}^{\text{future reward}} + b_{h,\delta}^t(s,a)\right\} \wedge 1. \tag{5.26}$$
> State value func.
> Set $\overline{V}_h^t(s) = \max_{a \in \mathcal{A}} \overline{Q}_h^t(s,a)$ and $\widehat{\pi}_h^t(s) = \arg\max_{a \in \mathcal{A}} \overline{Q}_h^t(s,a)$.
> Collect trajectory $\underbrace{(s_1^t, a_1^t, r_1^t), \ldots, (s_H^t, a_H^t, r_H^t)}_{\text{of that episode (t회차)}}$ according to $\widehat{\pi}^t$.

The UCB-VI algorithm will be analyzed using Lemma 15. In constructing functions $\overline{Q}_h$, we will need to satisfy two goals: (1) ensure that with high probability (5.22) is satisfied, i.e. $\overline{Q}_h$s are optimistic; and (2) that $\overline{Q}_h$s are "self-consistent," in the sense that the Bellman residuals in (5.23) are small. The second requirement already suggests that we should define $\overline{Q}_h$ approximately as a Bellman backup $\mathcal{T}_h^M \overline{Q}_{h+1}$, going backwards for $h = H+1, \ldots, 1$ as in dynamic programming, while ensuring the first requirement. In addition to these considerations, we will have to use a surrogate for the "Bellman operator" $\mathcal{T}_h^M$, since the model

대역 ↳ $\mathcal{T}_h^{\widehat{M}}$ 왜?

왜냐면 MDP를 정의하려면 transition prob.을 알아야하니깐..

Reward func: 알려져 있음.
ㅇㅎ · 모르는 것: Transition 확률!

$\hat{P}_h^t(s'|s,a) = \frac{n_h^t(s,a,s')}{n_h^t(s,a)}$

M is not known. This is achieved by estimating M (ㅇㅎ.) using empirical (estimated) transition frequencies. Putting these ideas together gives the update ("bonus" term) in (5.26). We apply the principle of value iteration, except that

1. For each episode t, we augment (강조!) the rewards $r_h(s, a)$ with a "bonus" $b^t_{h,\delta}(s, a)$ (designed to ensure optimism.)
2. The Bellman operator ($\mathcal{T}_h^M$) is approximated using the estimated transition probabilities in (5.25). ($\hat{P}_h^t(s'|s,a) = \frac{n_h^t(s,a,s')}{n_h^t(s,a)}$)

The bonus functions ($b_{h,\delta}(s,a)$) play precisely the same role as the width of the confidence interval (upper bound for the "instantaneous Regret") in (2.19): these bonuses ensure that (5.22) holds with high probability ($\bar{Q}_h$ s are optimistic), as we will show below in Lemma 16.

The following theorem shows that with an appropriate choice of bonus, this algorithm (UCB-VI) achieves a polynomial regret bound.

보너스 함수가 2 역할을 한다던데!

Theorem 1: For any $\delta > 0$, UCB-VI with

appropriate choice of bonus

$$b^t_{h,\delta}(s, a) = 2\sqrt{\frac{\log(2SAHT/\delta)}{n^t_h(s, a)}} \tag{5.27}$$

중요!

guarantees that with probability at least $1 - \delta$,

$$\mathbf{Reg} \lesssim HS\sqrt{AT} \cdot \sqrt{\log(SAHT/\delta)}$$

Regret 중요!

We mention that a slight variation on Lemma 18 below (using the Freedman inequality instead of the Azuma-Hoeffding inequality) yields an improved rate of $O(H\sqrt{SAT} + \mathrm{poly}(H, S, A)\log T)$, and the optimal rate can be shown to be $\Theta(\sqrt{HSAT})$; this is achieved through a more careful choice for the bonus $b^t_{h,\delta}$ and a more refined analysis. We remark that care should be taken in comparing results in the literature, as scaling conventions for the individual and cumulative rewards (as in Assumption 6) can vary.

5.6.1 Analysis for a Single Episode (within one episode.. t) (FIX t)

Our aim is to bound the regret

$$\mathbf{Reg} = \sum_{t=1}^{T} f^M(\pi_{M^\star}) - f^M(\pi^t)$$

for UCB-VI. To do so, we first prove several helper lemmas concerning the performance within each episode t. In what follows, we fix t ($\because$ for each episode t) and drop the superscript t.

Given the estimated transitions $\{\widehat{P}_h(\cdot \mid s, a)\}_{h,s,a}$, define the estimated MDP $\widehat{M} = \left\{\mathcal{S}, \mathcal{A}, \{\widehat{P}_h\}_{h=1}^H, \{R^M_h\}_{h=1}^H, d_1\right\}$. The associated "Bellman operator" is defined as..

언제 M이 뭔지 몰라서 정의 못했던

$\hat{P}_h^t(s'|s,a) = \frac{n_h^t(s,a,s')}{n_h^t(s,a)}$

$$\mathcal{T}^{\widehat{M}}_h Q(s, a) = r_h(s, a) + \mathbb{E}_{s'\sim\widehat{P}_h(\cdot|s,a)} \max_a Q(s', a) \tag{5.28}$$

for $Q : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. Consider the sequence of functions $\bar{Q}_h : \mathcal{S} \times \mathcal{A} \to [0, 1]$, $\bar{V}_h : \mathcal{S} \to [0, 1]$, for $h = 1, \ldots, H + 1$, (with $\bar{Q}_{H+1} \equiv 0$) and

$$\bar{Q}_h(s, a) = \left\{[\mathcal{T}^{\widehat{M}}_h \bar{Q}_{h+1}](s, a) + b_{h,\delta}(s, a)\right\} \wedge 1, \quad \text{and} \quad \bar{V}_h(s) = \max_a \bar{Q}_h(s, a) \tag{5.29}$$

for bonus functions $b_{h,\delta} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ to be chosen later. Henceforth, we follow the usual notation that for functions f, g over the same domain, $f \leq g$ indicates pointwise inequality over the domain.

upper

The first lemma we present shows that as long as the bonuses $b_{h,\delta}$ are large enough to bound the error (between the estimated transition probabilities and true transition probabilities,) the functions $\overline{Q}_1, \ldots, \overline{Q}_H$ constructed above will be optimistic.

$\ast$ ↳ $\overline{Q}_h \geq Q_h^{M,*}$

그냥 그대로 수식으로만.

transition probability ; bonus

Lemma 16: Suppose we have estimates $\{\widehat{P}_h(\cdot \mid s,a)\}_{h,s,a}$ and a function $b_{h,\delta} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ with the property that for all $s \in \S, a \in \mathcal{A}$,

error b.w. ; bonuses Large Enough

$$\left| \sum_{s'} \widehat{P}_h(s' \mid s,a) V_h^{M,\star}(s') - \sum_{s'} P_h^M(s' \mid s,a) V_h^{M,\star}(s') \right| \leq b_{h,\delta}(s,a). \tag{5.30}$$

estimated trans. prob ; true trans. prob

Then for all $h \in [H]$, we have

1

$$\overline{Q}_h \geq Q_h^{M,\star}, \quad \text{and} \quad \overline{V}_h \geq V_h^{M,\star} \tag{5.31}$$

↳ optimism! (over estimate가 guarantee 됨)

for $\overline{Q}_h, \overline{V}_h$ defined in (5.29).

Proof of Lemma 16. The proof proceeds by backward induction on the statement

$$\overline{V}_h \geq V_h^{M,\star}$$

with $h = H+1$ down to $h = 1$. We start with the (1) base case $h = H+1$, which is trivial because $\overline{V}_{H+1} = V_{H+1}^{M,\star} \equiv 0$. Now, (2) assume $\overline{V}_{h+1} \geq V_{h+1}^{M,\star}$, and let us (3) prove the induction step. Fix $(s,a) \in \mathcal{S} \times \mathcal{A}$. If $\overline{Q}_h(s,a) = 1$, then, trivially, $\overline{Q}_h(s,a) \geq Q_h^{M,\star}(s,a)$. Otherwise, $\overline{Q}_h(s,a) = \mathcal{T}_h^{\widehat{M}} \overline{Q}_{h+1}(s,a) + b_{h,\delta}(s,a)$, and thus

↳ [0,1] 가정해서 당연. oo.

$$\begin{aligned}
\overline{Q}_h(s,a) - Q_h^{M,\star}(s,a) &\overset{\text{정의}}{=} b_{h,\delta}(s,a) + \mathbb{E}_{s' \sim \widehat{P}_h(\cdot|s,a)} \overline{V}_{h+1}(s') - \mathbb{E}_{s' \sim P_h^M(\cdot|s,a)} V_{h+1}^{M,\star}(s') \\
&\geq b_{h,\delta}(s,a) + \mathbb{E}_{s' \sim \widehat{P}_h(\cdot|s,a)} V_{h+1}^{M,\star}(s') - \mathbb{E}_{s' \sim P_h^M(\cdot|s,a)} V_{h+1}^{M,\star}(s') \geq 0.
\end{aligned}$$

커야됨. ; by assumption (2) ; → $\overline{Q}_h \geq Q_h^{M,*}$

← 위 식의 결과

This, in turn, implies that $\overline{V}_h(s) \underset{\text{정의}}{=} \max_a \overline{Q}_h(s,a) \geq \max_a Q_h^{M,\star}(s,a) \underset{\text{정의}}{=} V_h^{M,\star}(s)$, concluding the induction step. oo. $\square$

We now analyze the effect of using an estimated model $\widehat{M}$ for the Bellman operator (rather than the true unknown $\mathcal{T}_h^M$.)

모르는데 이게서 그래서 안 쓰지

transition probability

Lemma 17: Suppose we have estimates $\{\widehat{P}_h(\cdot \mid s,a)\}_{h,s,a}$ and $b'_{h,\delta}(s,a) : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ with the property that

$$\max_{V \in \{0,1\}^{\mathcal{S}}} \left| \sum_{s'} \widehat{P}_h(s' \mid s,a) V(s') - \sum_{s'} P_h^M(s' \mid s,a) V(s') \right| \leq b'_{h,\delta}(s,a) \tag{5.32}$$

$\max_{V \in [0,1]^S} \left| \mathbb{E}_{s' \sim \widehat{P}_h(\cdot|s,a)} V(s') - \mathbb{E}_{s' \sim P_h^M(\cdot|s,a)} V(s') \right|$ oo.

↓ 강 증명 막판에 쓰이는 가정.

Then the Bellman residual satisfies

2

$$\overline{Q}_h - \mathcal{T}_h^M \overline{Q}_{h+1} \leq (b_{h,\delta} + b'_{h,\delta}) \wedge 1. \tag{5.33}$$

for $\bar{Q}_h, \bar{V}_h$ defined in (5.29).

Proof of Lemma 17. That $\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1} \underset{\infty}{\leq} 1$ is immediate. To prove the main result, observe that

$$\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1} \overset{\text{def.}}{=} \left\{ \mathcal{T}_h^{\widehat{M}} \bar{Q}_{h+1} + b_{h,\delta} \right\} \wedge 1 - \mathcal{T}_h^M \bar{Q}_{h+1} \overset{1..}{\leq} (\mathcal{T}_h^{\widehat{M}} - \mathcal{T}_h^M) \bar{Q}_{h+1} + b_{h,\delta} \tag{5.34}$$

$\leq b'_{h,\delta}$ 시. ⇒ 결론!

For any $Q \in \mathcal{S} \times \mathcal{A} \to [0, 1]$,

$$(\mathcal{T}_h^{\widehat{M}} - \mathcal{T}_h^M) Q(s, a) \overset{\text{def}}{=} \mathbb{E}_{s' \sim \widehat{P}_h(\cdot|s,a)} \max_a Q(s', a) - \mathbb{E}_{s' \sim P_h^M(\cdot|s,a)} \max_a Q(s', a) \tag{5.35}$$

$$\leq \max_{V \in [0,1]^S} |\mathbb{E}_{s' \sim \widehat{P}_h(\cdot|s,a)} V(s') - \mathbb{E}_{s' \sim P_h^M(\cdot|s,a)} V(s')|. \tag{5.36}$$

$\leq b'_{h,\delta}(s,a)$ ⇐ (5.32)의 "given property"

Since the maximum is achieved at a vertex of $[0, 1]^S$, the statement follows. $\square$

$\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1} \leq (b_{h,\delta} + b'_{h,\delta})$ 시

5.6.2 Regret Analysis

$\widehat{P}_h(\cdot|s,a)$

We now bring back the time index t and show that the estimated transition probabilities in UCB-VI satisfy conditions of Lemma 16 and Lemma 17, ensuring that the functions $\bar{Q}_1^t, \ldots, \bar{Q}_H^t$ are optimistic.

estimated transition probabilities

Lemma 18: Let $\{\widehat{P}_h^t\}_{h \in [H], t \in [T]}$ be defined as in (5.25). Then with probability at least $1 - \delta$, the functions

$$b_{h,\delta}^t(s, a) = 2\sqrt{\frac{\log(2SAHT/\delta)}{n_h^t(s, a)}}, \quad \text{and} \quad b_{h,\delta}'^t(s, a) = 8\sqrt{\frac{S \log(2SAHT/\delta)}{n_h^t(s, a)}}$$

satisfy the assumptions of Lemma 16 and Lemma 17, respectively, for all $s \in \mathcal{S}$, $a \in \mathcal{A}$, $h \in [H]$, and $t \in [T]$ simultaneously.

Proof of Lemma 18. We leave the proof as an exercise. $\square$

Proof of Theorem 1. Putting everything together, we can now prove Theorem 1. Under the event in Lemma 18, the functions $\bar{Q}_1^t, \ldots, \bar{Q}_H^t$ are optimistic, which means that the conditions of Lemma 15 hold, and the instantaneous regret on round t (conditionally on $s_1 \sim d_1$) is at most

↳ Reg: $V_1^{M,*}(s) - V_1^{M,\widehat{\pi}}(s)$

RHS of Lemma 15

$$\sum_{h=1}^H \mathbb{E}^{M,\widehat{\pi}^t} \left[(\bar{Q}_h^t - \mathcal{T}_h^M \bar{Q}_{h+1}^t)(s_h^t, \widehat{\pi}_h^t(s_h^t)) \mid s_1 = s \right] \leq \sum_{h=1}^H \mathbb{E}^{M,\widehat{\pi}^t} \left[(b_{h,\delta}(s_h^t, \widehat{\pi}_h^t(s_h^t)) + b'_{h,\delta}(s_h^t, \widehat{\pi}_h^t(s_h^t))) \wedge 1 \right],$$

first inequality: Lemma 15

where the second inequality invokes Lemma 17. Summing over $t = 1, \ldots, T$, and applying the Azuma-Hoeffding inequality, we have that with probability at least $1 - \delta$, the regret of UCB-VI is bounded by

Summing over $t=1,\cdots,T$

$$\sum_{t=1}^T \sum_{h=1}^H \mathbb{E}^{M,\widehat{\pi}^t} \left[(b_{h,\delta}(s_h^t, \widehat{\pi}_h^t(s_h^t)) + b'_{h,\delta}(s_h^t, \widehat{\pi}_h^t(s_h^t))) \wedge 1 \right]$$

$$\lesssim \sum_{t=1}^T \sum_{h=1}^H (b_{h,\delta}(s_h^t, \widehat{\pi}_h^t(s_h^t)) + b'_{h,\delta}(s_h^t, \widehat{\pi}_h^t(s_h^t))) \wedge 1 + \sqrt{HT \log(1/\delta)}.$$

Azuma-Hoeffding

<Lemma 15>

appropriate choice of bonus

$$b^t_{h,\delta}(s,a) = 2\sqrt{\frac{\log(2SAHT/\delta)}{n^t_h(s,a)}}$$ 로!

∞ 이렇게 정의됨.

Using the bonus definition in (5.27), the bonus term above is bounded by 왼쪽

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\sqrt{\frac{S\log(2SAHT/\delta)}{n^t_h(s^t_h,\widehat{\pi}^t_h(s^t_h))}}\wedge 1 \le \sqrt{S\log(2SAHT/\delta)}\sum_{t=1}^{T}\sum_{h=1}^{H}\frac{1}{\sqrt{n^t_h(s^t_h,\widehat{\pi}^t_h(s^t_h))}}\wedge 1 \quad (5.37)$$

리미계도 ≤

The double summation can be handled in the same fashion as Lemma 8:

$$\sum_{t=1}^{T}\sum_{h=1}^{H}\frac{1}{\sqrt{n^t_h(s^t_h,\widehat{\pi}^t_h(s^t_h))}}\wedge 1 = \sum_{h=1}^{H}\sum_{(s,a)}\sum_{t=1}^{T}\frac{\mathbb{I}\{(s^t_h,\widehat{\pi}^t_h(s^t_h))=(s,a)\}}{\sqrt{n^t_h(s,a)}}\wedge 1$$

$$\lesssim \underbrace{\sum_{h=1}^{H}}_{H}\underbrace{\sum_{(s,a)}\sqrt{n^T_h(s,a)}}_{\sqrt{SAT}} \le H\sqrt{SAT}.$$

*Lemma 8 pf에서...

$$\sum_{t=1}^{n}\frac{1}{\sqrt{t-1}}\wedge 1 \le 1+2\sqrt{n}$$

⋯> Regret is Bounded

□

6. GENERAL DECISION MAKING

So far, we have covered three general frameworks for interaction decision making: The contextual bandit problem, the structured bandit problem, and the episodic reinforcement learning problem; all of these frameworks generalize the classical multi-armed bandit problem in different directions. In the context of structured bandits, we introduced a complexity measure called the Decision-Estimation Coefficient (DEC), which gave a generic approach to algorithm design, and allowed us to reduce the problem of interactive decision making to that of supervised online estimation. In this section, we will build on this development on two fronts: First, we will introduce a unified framework for decision making, which subsumes all of the frameworks we have covered so far. Then, we will show that i) the Decision-Estimation Coefficient and its associated meta-algorithm (E2D) extend to the general decision making framework, and ii) boundedness of the DEC is not just sufficient, but actually *necessary* for low regret, and thus constitutes a fundamental limit. As an application of the general tools we introduce, we will show how to use the (generalized) Decision-Estimation Coefficient to solve the problem of tabular reinforcement learning (Section 6.6), offering an alternative to the UCB-VI method we introduced in Section 5.

6.1 Setting

For the remainder of the course, we will focus on a framework called Decision Making with Structured Observations (DMSO), which subsumes all of the decision making frameworks we have encountered so far. The protocol proceeds in T rounds, where for each round $t = 1, \ldots, T$:

1. The learner selects a *decision* $\pi^t \in \Pi$, where Π is the *decision space*.
2. Nature selects a *reward* $r^t \in \mathcal{R}$ and *observation* $o^t \in \mathcal{O}$ based on the decision, where $\mathcal{R} \subseteq \mathbb{R}$ is the *reward space* and $\mathcal{O}$ is the *observation space*. The reward and observation are then observed by the learner.

We focus on a stochastic variant of the DMSO framework.